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Semiconductor fabrication apparatus and fabrication method

Abstract

A semiconductor fabrication apparatus comprises a process chamber, an ozone supply that provides the process chamber with ozone, an oxygen supply that provides the ozone supply with a source gas of the ozone, and a plurality of impurity detectors disposed between the oxygen supply and the ozone supply. The impurity detectors detect an inactive gas in the source gas.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This U.S. nonprovisional application claims priority under 35 U.S.C § 119 from Korean Patent Application No. 10-2021-0104594, filed on Aug. 9, 2021 in the Korean Intellectual Property Office, the contents of which are herein incorporated by reference in their entirety.

TECHNICAL FIELD

(2) Embodiments of the present inventive, concepts are directed to semiconductor fabrication

apparatuses and fabrication methods, and more particularly, to semiconductor fabrication apparatuses and fabrication methods in which a thin film is formed on substrate.

DISCUSSION OF THE RELATED ART

(3) In general, the fabrication of a semiconductor device requires a plurality of processes such as deposition, photolithography, and cleaning. Among these processes, deposition processes, such as chemical vapor deposition (CVD) or atomic layer deposition (ALD), are used to form a material layer on a substrate.

SUMMARY

(4) Some embodiments of the present inventive concepts provide semiconductor fabrication apparatuses that increase a production yield of thin films.

(5) According to some embodiments of the present inventive concepts, a semiconductor fabrication apparatus includes: a process chamber; an ozone supply that provides the process chamber with ozone; an oxygen supply that provides the ozone supply with a source gas for the ozone; and a plurality of impurity detectors disposed between the oxygen supply and the ozone supply. The impurity detectors detect an inactive gas in the source gas.

(6) According to some embodiments of the present inventive concepts, a semiconductor fabrication apparatus includes: an ozone supply that provides ozone; an oxygen supply that provides the ozone supply with a source gas for the ozone; an oxygen purifier disposed between the oxygen supply and the ozone supply, where the oxygen purifier purifies the source gas; a first impurity detector disposed between the oxygen purifier and the oxygen supply, where the first impurity detector detect carbon dioxide in the source gas; and a second impurity detector disposed between the oxygen purifier and the ozone supply, where the second impurity detector detects an inactive gas in the source gas.

(7) According to some embodiments of the present inventive concepts, a method of fabricating a semiconductor includes: loading a substrate onto a chuck; and forming a thin film on the substrate. The step of forming the thin film includes: supplying, a source gas; purifying the source gas by removing carbon dioxide from the source gas; and detecting an inactive gas in the source gas.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram of a semiconductor fabrication apparatus according to some embodiments of the present inventive concepts.

(2) FIG. 2 is a cross-sectional view of an example of an impurity detector shown in FIG. 1.

(3) FIG. 3 is a cross-sectional view of an example of a gas cell shown in FIG. 2.

(4) FIG. 4 is a perspective view of an example of an optical fiber shown in FIG. 2.

(5) FIG. 5 is a cross-sectional view of an example of an optical sensor shown in FIG. 2.

(6) FIG. 6 is a flow chart of a semiconductor fabrication method according to some embodiments of the present inventive concepts.

(7) FIG. 7 is a flow chart of a thin-film formation step shown in FIG. 6.

DETAILED DESCRIPTION

(8) FIG. 1 shows a semiconductor fabrication apparatus **100** according to embodiments of the present inventive concepts.

(9) Referring to FIG. 1, the semiconductor fabrication apparatus **100** according to an embodiment of the present inventive concepts includes a chemical vapor deposition apparatus. Alternatively, the semiconductor fabrication apparatus **100** according to an embodiment of the present inventive concepts includes a physical vapor deposition apparatus, but embodiments of the present inventive concepts are not limited thereto. In an embodiment, the semiconductor fabrication apparatus **100** includes a process chamber **10**, an ozone supply **20**, an oxygen supply **30**, an oxygen purifier **40**,

and impurity detectors **50**.

(10) The process chamber **10** provides a substrate **W** with a hermetically sealed space to perform a fabrication process on the substrate **W**. The fabrication process may include a thin-film deposition process. For example, the fabrication process may be chemical vapor deposition (CVD) or physical vapor deposition (PVD), but embodiments of the present inventive concepts are not limited thereto. The process chamber **10** includes, for example, a chuck **12** and a showerhead **14**.

(11) The chuck **12** is disposed in a lower portion of the process chamber **10**. The chuck **12** can load the substrate **W**. The chuck **12** uses a first radio frequency (RF) power **18** provided by a first power supply **16** to induce a plasma **P** on the substrate **W**. The chuck **12** may be one of an electrostatic chuck, a heater plate, or a susceptor, but embodiments of the present inventive concepts are not limited thereto.

(12) The showerhead **14** is disposed in an upper portion of the process chamber **10**, above the chuck **12**. The showerhead **14** is connected to a precursor supply **13** and the ozone supply **20**. The showerhead **14** provides the substrate **W** with a precursor gas **15** received from precursor supply **13** and ozone **34** received from the ozone supply **20** and forms a thin film on the substrate **W**. For example, thin film includes a metal oxide such as zirconium oxide (ZrO_2). For another example, the thin film includes silicon oxide (SiO_2), but embodiments of the present inventive concepts are not limited thereto.

(13) The precursor supply **13** is located on one side of the showerhead **14**. The precursor supply **13** provides the showerhead **14** with the precursor gas **15**. The precursor gas **15** includes a zirconium compound. Additionally or alternatively, the precursor gas **15** may include silane, but embodiments of the present inventive concepts are not limited thereto.

(14) The first power supply **16** is connected to the chuck **12**. In addition, the first power supply **16** is connected to the showerhead **14**, but embodiments of the present inventive concepts are not limited thereto. The first power supply **16** provides the chuck **12** or the showerhead **14** with the first RF power **18** and produces the plasma **P** from the precursor gas **15** and the ozone **34**. The first RF power **18** ranges from about 1 kW to about 10 kW. The first RF power **18** has a pulse frequency of about 400 kHz to about 60 MHz.

(15) The ozone supply **20** is connected between the process chamber **10** and the oxygen purifier **40**. The ozone supply **20** is located on another side of the showerhead **14**, opposite from the precursor supply **13**. The ozone supply **20** provides the showerhead **14** with the ozone **34**. The ozone supply **20** includes first electrodes **22**. The first electrodes **22** use a second RF power **28** received from second power supply **26** and produces the ozone **34** from a source gas **32**. The source gas **32** for the ozone **34** includes oxygen (O_2).

(16) The second power supply **26** is connected to the first electrodes **22**. The second power supply **26** provides the first electrodes **22** with the second RF power **28**. The second RF power **28** is less than the first RF power **18**. The second RF power **28** ranges from about 100 W to about 1 kW.

(17) The oxygen supply **30** is connected to the oxygen purifier **40** and the ozone supply **20**. The oxygen supply **30** provides the oxygen purifier **40** with the source gas **32**. The oxygen supply **30** includes an oxygen storage tank. The source gas **32** includes oxygen, carbon dioxide, and an inactive or inert gas, referred to hereinafter as the inactive gas.

(18) The oxygen purifier **40** is connected between the oxygen supply **30** and the ozone supply **20**. The oxygen purifier **40** adsorbs the carbon dioxide (CO_2) in the source gas **32** and purifies the oxygen of the source gas **32**. In addition, the oxygen purifier **40** adsorbs organic materials in the source gas **32**. Furthermore, the oxygen purifier **40** includes an adsorption catalyst, such as platinum, but embodiments of the present inventive concepts are not limited thereto.

(19) At least one of the impurity detectors **50** is disposed between the ozone supply **20** and the oxygen purifier **40**. In addition, another impurity detector **50** is disposed between the oxygen supply **30** and the oxygen purifier **40**. The impurity detectors **50** detect the inactive gas, such as N_2 or Ar, in the source gas **32**. The inactive gas can cause a failure of thin-film deposition.

When the impurity detectors **50** detect the inactive gas in the source gas **32**, a controller generates an interlock control signal that reduces or minimizes failure of thin-film deposition performed in the process chamber **10**. The oxygen supply **30** is replaced with a new oxygen supply. Therefore, the semiconductor fabrication apparatus **100** according to an embodiment of the present inventive concepts uses the impurity detectors **50** to increase a production yield of thin films.

(20) In an embodiment, the impurity detectors **50** include a first impurity detector **52** and a second impurity detector **54**.

(21) The first impurity detector **52** is disposed between the oxygen supply **30** and the oxygen purifier **40**. The first impurity detector **52** detects carbon dioxide. Alternatively, in an embodiment, the first impurity detector **52** detects the inactive gas, but embodiments of the present inventive concepts are not limited thereto.

(22) The second impurity detector **54** is disposed between the ozone supply **20** and the oxygen purifier **40**. The second impurity detector **54** detects the inactive gas after removal of the carbon dioxide.

(23) Accordingly, the controller uses detection signals of the first and second impurity detectors and **54** to determine a removal amount of the carbon dioxide in the oxygen purifier **40**.

(24) Third power supplies **56** are connected to the first impurity detector **52** and the second impurity detector **54**. The third power supplies **56** provide the first and second impurity detectors **52** and **54** with third RF powers **58** that induce a plasma reaction between the oxygen, the carbon dioxide, and the inactive gas of the source gas **32**. The third RF power **58** is less than the second RF power **28**. For example, the third RF power **58** ranges from about 10 W to about 100 W. The third RF power **58** has a frequency of from about 20 kHz to about 60 kHz and an alternating current (AC) of from about 1 kV to about 7 kV. The first impurity detector **52** and the second impurity detector **54** detect plasma light (see **61** of FIG. 2) of the carbon dioxide and the inactive gas.

(25) FIG. 2 shows an example of the first impurity detector **52** and the second impurity detector **54** depicted in FIG. 1.

(26) Referring to FIG. 2, in an embodiment, each of the first impurity detector **52** and the second impurity detector **54** includes a gas cell **62**, second electrodes **64**, an optical system **68**, an optical fiber **72**, a spectroscope **74**, and an optical sensor **76**.

(27) The gas cell **62** stores the source gas **32**. The gas cell **62** includes a view port **66**. The plasma light **61** of the source gas **32** is externally exposed outside of the gas cell **62** through the view port **66**.

(28) The second electrode **64** provides the gas cell **62** with high degree of airtightness. The second electrodes **64** are connected to the third power supply **56**. A resistor **57** is connected between the third power supply **56** and one of the second electrodes **64**. The resistor **57** adjusts the third RF power **58**. The second electrodes **64** use the third RF power **58** to generate the plasma light **61** of the source gas **32**. In a plasma reaction, oxygen in the source gas **32** produces the plasma light **61** with various discontinuous emission bands with a wide bandwidth that ranges between ultraviolet and near-infrared. Nitrogen (N.sub.2) and argon (Ar) in the source gas **32** generate the discontinuous emission bands. For example, nitrogen (N.sub.2) generates plasma light **61** with a wavelength of about 358 nm, and argon (Ar) generates plasma light **61** with a wavelength of about 813 nm. When the source gas **32** includes carbon dioxide or an organic material, the source gas **32** generates plasma light **61** at specific plasma emission bands from molecular and constituent atoms of the carbon dioxide or the organic material. The first impurity detector **52** and the second impurity detector **54** detect the plasma light **61** and analyze a concentration of the impurities. When analyzing an extremely small amount of impurities, such as nitrogen (N.sub.2), argon (Ar), carbon dioxide, and an organic material, present in the source gas **32**, the controller uses the intensity of oxygen emission band signals to normalize the intensity of the plasma light **61** and increase the accuracy of the impurity analysis.

(29) FIG. 3 shows an example of the gas cell **62** depicted in FIG. 2.

(30) Referring to FIG. 3, in an embodiment, the gas cell **62** is a glow discharge gas cell. The second electrodes **64** include a cathode and an anode, and have a thickness of about 1 mm and a space therebetween. The second electrodes **64** include gold (Au), but embodiments of the present inventive concepts are not limited thereto. Each of the second electrodes **64** has a rounded shape at its tip. The gas cell **62** is provided with the source gas **32** in opposite directions. For example, the source gas **32** is introduced in an upper direction into the gas cell **62** and is discharged in a downward direction from the gas cell **62**. In an embodiment of the present inventive concepts, because the analysis targets are nitrogen (N.sub.2), argon (Ar), carbon dioxide (CO.sub.2), etc., or main components of air, the gas cell **62** should be airtight to exclude interference of external air, and to accomplish the airtightness, special attention is needed when installing the electrodes, the view port **66**, etc. A pressure range of the source gas **32** between about 30 Torr and about 80 Torr is an optimum condition in plasma emission analysis. It is possible not only to maintain a high density of plasma but also to maintain a low intensity baseline signal, which results in a maximum increase in analysis sensitivity.

(31) Referring back to FIG. 2, in an embodiment, the optical system **68** is located on the view port **66**. The optical system **68** provides the optical fiber **72** with the plasma light **61** of the source gas **32**. The optical system **68** includes a first lens **65**, a filter **67**, and a second lens **69**.

(32) The first lens **65** is disposed between the view port **66** and the filter **67**. The first lens **65** projects the plasma light **61** of the source gas **32** onto the filter **67**. The first lens **65** includes a convex lens, but embodiments of the present inventive concepts are not limited thereto.

(33) The filter **67** is disposed between the first lens **65** and the second lens **69**. The filter **67** selectively transmits or reflects a specific wavelength range chosen from a range between ultraviolet wavelength and near-infrared wavelength of the plasma light **61** that is a measurement target.

(34) The second lens **69** is disposed between the filter **67** and the optical fiber **72**. The second lens **69** focuses plasma light **61** on the optical fiber **72**. The second lens **69** includes a convex lens. The second lens **69** may include an aspherical lens, but embodiments of the present inventive concepts are not limited thereto.

(35) The optical fiber **72** is disposed between the second lens **69** and the spectroscope **74**. The optical fiber **72** transmits the plasma light **61** of the carbon dioxide and the inactive gas to the spectroscope **74**.

(36) FIG. 4 shows an example of the optical fiber **72** depicted in FIG. 2.

(37) Referring to FIG. 4, in an embodiment, the optical fiber **72** includes a plurality of optical fibers **72**. The plurality of optical fibers **72** are coupled into a bundle. An input end **721** of the optical fibers **72** has a circular bundle shape, and maximally increases a light harvesting efficiency of the plasma light **61**. An output end **722** of the optical fibers **72** has a linear bundle shape and is linearly arranged to maximally increase resolution of the spectroscope **74**.

(38) Referring again to FIG. 2, in an embodiment, the spectroscope **74** is connected between the optical fiber **72** and the optical sensor **76**. The spectroscope **74** may include a prism or a diffraction grating.

(39) The optical sensor **76** is disposed adjacent to the spectroscope **74**. The optical sensor **76** detects the plasma light **61** of the carbon dioxide and the inactive gas. For example, in an embodiment, the optical sensor **76** includes a sensor array of complementary metal oxide semiconductor (CMOS) sensors or charge-coupled devices (CCD). Alternatively, in an embodiment, the optical sensor **76** may include a photodiode, but embodiments of the present inventive concepts are not limited thereto. In an embodiment, optical sensor **76** includes a pixel whose width is less than a width or thickness of the optical fiber **72**. In addition, the optical sensor **76** includes a pixel array whose height or length is greater than a vertical height of the output end **722** of the optical fiber **72**.

(40) FIG. 5 shows an example of the optical sensor **76** depicted in FIG. 2.

(41) Referring to FIG. 5, in an embodiment, the optical sensor **76** is cooled by a cooler **82**. The

cooler **82** is in contact with one side or a bottom surface of the optical sensor **76**. The cooler **82** includes a thermoelectric semiconductor device. The cooler **82** is provided on a heat sink **84**. A temperature of the cooler **82** is controlled by a cooling controller **78**. For example, the optical sensor **76** maintained at room temperature, e.g. about 20° C.

(42) Accordingly, the semiconductor fabrication apparatus **100** according to an embodiment of the present inventive concepts uses the first impurity detector **52** and the second impurity detector **54** to detect the inactive gas in the source gas **32** during fabrication processes performed on the substrate W, thereby increasing a production yield of thin films.

(43) The following will describe a semiconductor fabrication method according to embodiments of the present inventive concepts that uses the semiconductor fabrication apparatus **100**.

(44) FIG. **6** shows an example of a semiconductor fabrication method according to an embodiment of the present inventive concepts.

(45) Referring to FIGS. **1** and **6**, in an embodiment, the chuck **12** loads the substrate W (**S10**). A robot arm loads the substrate W onto the chuck **12** in the process chamber **10**. The chuck **12** uses an electrostatic voltage to hold the substrate W. Alternatively, in an embodiment, the chuck **12** heats the substrate W, but embodiments of the present inventive concepts are not limited thereto.

(46) Afterwards, the process chamber **110** uses the ozone **34** and the precursor gas **15** to form a thin film on the substrate W (**S20**). For example, the thin film may include zirconium oxide (ZrO.sub.2) or silicon oxide (SiO.sub.2).

(47) FIG. **7** shows an example of the thin-film formation step **S20** depicted in FIG. **6**.

(48) Referring to FIGS. **1** and **7**, in an embodiment, the oxygen supply **30** supplies the source gas **32** (**S210**). The source gas **32** has a flow rate of about 1 SCCM to about 1,000 SCCM.

(49) The first impurity detector **52** detects carbon dioxide in the source gas **32** (**S220**). The first impurity detector **52** uses the third RF power **58** to induce a plasma reaction in the source gas **32** and optically detects the carbon dioxide in the source gas **32**. The optical sensor **76** of the first impurity detector **52** detects the plasma light **61** of the carbon dioxide. In addition, the first impurity detector **52** may detect the inactive gas in the source gas **32**, but embodiments of the present inventive concepts are not limited thereto.

(50) The oxygen purifier **40** removes the carbon dioxide from the source gas **32** to purify the oxygen of the source gas **32** (**S230**). The carbon dioxide is adsorbed by a catalyst in the oxygen purifier **40**. In addition, the oxygen purifier **40** removes organic material from the source gas **32** to purify the source gas **32**.

(51) The second impurity detector **54** detects the inactive gas in the source gas **32** (**S240**). The second impurity detector **54** uses the third RF power **58** to induce a plasma reaction in the source gas **32** to optically detect the inactive gas in the source gas **32**. The optical sensor **76** of the second impurity detector **54** detects the plasma light **61** of the inactive gas.

(52) The controller uses a detection signal of the second impurity detector **54** to determine whether or not the inactive gas is present in the source gas **32** (**S250**).

(53) When the inactive gas is absent in the source gas **32**, the ozone supply **20** produces the ozone **34** from the source gas **32** and supplies the showerhead **14** with the ozone **34** (**S260**). The second power supply **26** provides the second RF power **28** to the first electrodes **22** of the ozone supply **20**, and the source gas **32** is excited into a plasma state to generate the ozone **34**. The ozone **34** is provided to the showerhead **14**. The precursor supply **13** supplies the showerhead **14** with the precursor gas **15**. The showerhead **14** provides the substrate W with the ozone **34** and the precursor gas **15**.

(54) The first power supply **16** supplies the chuck **12** with the first RIF power **18** to generate the plasma P from the ozone **34** and the precursor gas **15** (**S270**). The ozone **34** and the precursor gas **15** deposit a thin film on the substrate W. The plasma P increases a chemical reaction between the ozone **34** and the precursor gas **15** and increases homogeneity and/or uniformity of the thin film.

(55) When the inactive gas is present in the source gas **32**, the controller outputs an interlock

control signal (S280). When the interlock control signal is output, a thin-film deposition process of the substrate W is terminated in the process chamber 10, with the result that a failure of thin-film deposition can be prevented or minimized.

(56) Accordingly, a semiconductor fabrication method according to embodiments of the present inventive concepts detects an inactive gas in the source gas 32 and increases a production yield of thin films.

(57) As described above, a semiconductor fabrication apparatus according to embodiments of the present inventive concepts uses impurity detectors to detect an inactive gas in a source gas that is used in a substrate fabrication process to increase a production yield of thin films.

(58) The above descriptions are specific examples for practicing the present inventive concepts. Embodiments of the present inventive concepts include not only embodiments described above but also embodiments that can be easily or simply changed in design. In addition, embodiments of the present inventive concepts also include techniques that can be easily modified and implemented using embodiments described above.

Claims

1. A semiconductor fabrication apparatus, comprising: a process chamber that includes a chuck and a first power supply connected to the chuck; an ozone supply that provides the process chamber with ozone, wherein the ozone supply includes a first supply line connected to the process chamber, first electrodes disposed on both sides of the first supply line, and second power supply connected to the first electrodes; an oxygen supply that provides the ozone supply with a source gas for the ozone; and a plurality of impurity detectors disposed between the oxygen supply and the ozone supply, wherein the impurity detectors detect an inactive gas in the source gas, each of the plurality of impurity detectors includes a gas cell, second electrodes disposed in the gas cell, and third power supplies connected to the second electrodes, wherein the second electrodes are smaller than the first electrodes, wherein the first power supply provides a first radio frequency power into the chuck to generate a first plasma of a process gas, wherein the second power supply provides a second radio frequency power less than the first radio frequency power into the first electrodes to generate the ozone from the source gas, wherein each of the third power supplies provides a third radio frequency power less than the second radio frequency power into the second electrodes to generate a second plasma of the source gas, wherein the first radio frequency power is 1 KW to 10 KW, wherein the second radio frequency power is 100 W to 1 KW, and wherein the third radio frequency power is 10 W to 100 W.

2. The apparatus of claim 1, further comprising an oxygen purifier disposed between the oxygen supply and the ozone supply, wherein the oxygen purifier removes carbon dioxide in the source gas.

3. The apparatus of claim 2, wherein the impurity detectors include: a first impurity detector disposed between the oxygen supply and the oxygen purifier, wherein the first impurity detector detects the carbon dioxide; and a second impurity detector disposed between the oxygen purifier and the ozone supply, wherein the second impurity detector detects the inactive gas.

4. The apparatus of claim 1, wherein each of the third power supplies each include a resistor configured to adjust the third radio frequency power.

5. The apparatus of claim 1, wherein each of the impurity detectors further includes: an optical system disposed on a view port of the gas cell, wherein the optical system projects plasma light of the carbon dioxide or the inactive gas; and an optical sensor connected to the optical system, wherein the optical sensor detects the plasma light.

6. The apparatus of claim 5, wherein each of the impurity detectors further includes: an optical fiber disposed between the optical system and the optical sensor; and a spectroscope disposed between the optical fiber and the optical sensor.

7. The apparatus of claim 6, wherein the optical fiber includes a plurality of optical fibers, wherein the plurality of optical fibers include: an input end that has a circular bundle shape; and an output end that has a linear bundle shape.

8. The apparatus of claim 5, wherein each of the impurity detectors further includes a cooler in contact with the optical sensor, wherein the cooler cools the optical sensor.

9. The apparatus of claim 1, wherein the second electrodes include gold (Au).
